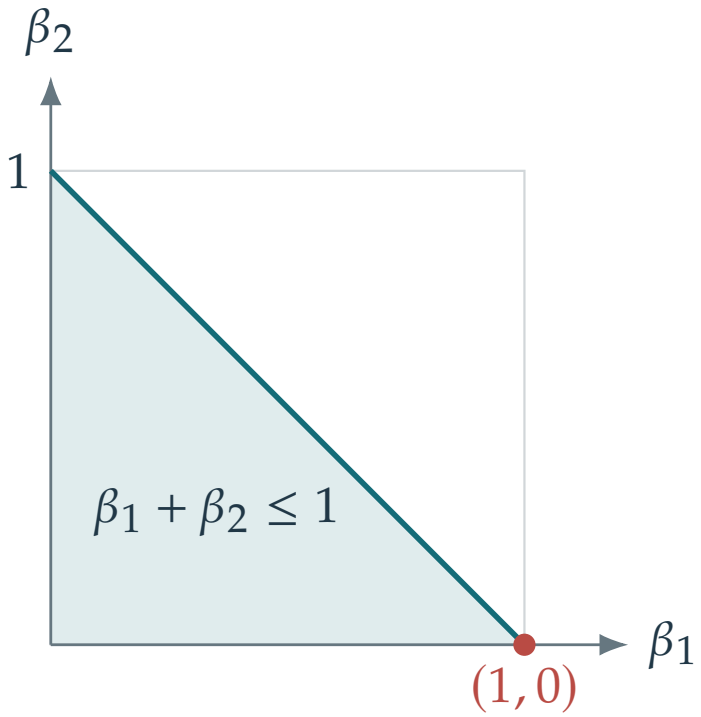
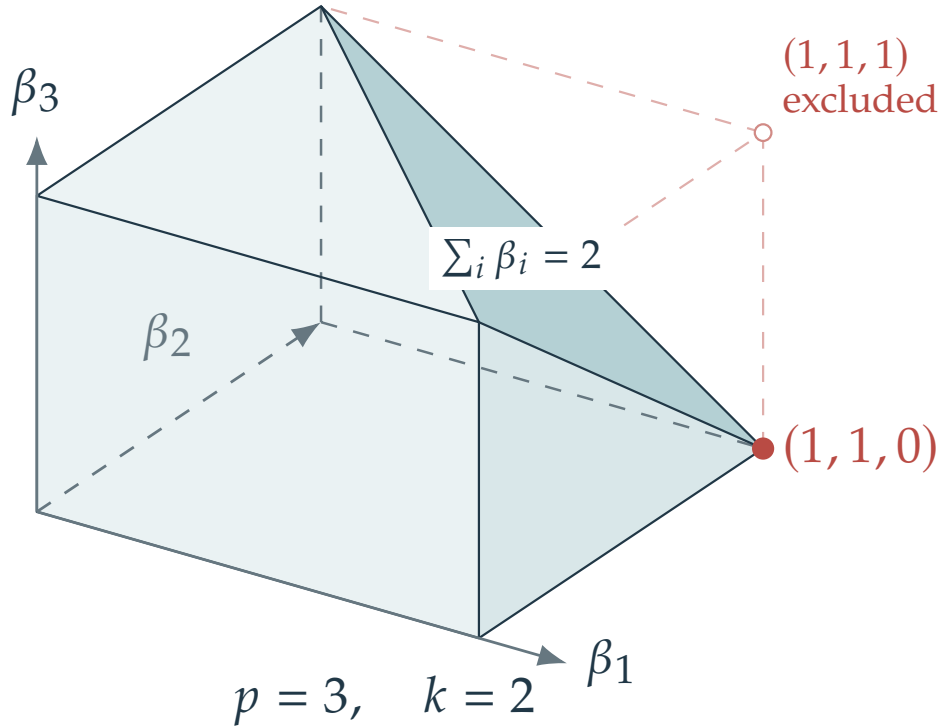


$p = 2, \quad k = 1$



$p = 3, \quad k = 2$

Feasible region: $0 \le \beta_i \le 1, \sum_i \beta_i \le k$. Red dots: a maximizing vertex.

$$\lambda_1 \ge \cdots \ge \lambda_p \ge 0, \qquad \max \sum_i \lambda_i \beta_i = \sum_{i=1}^{k} \lambda_i$$